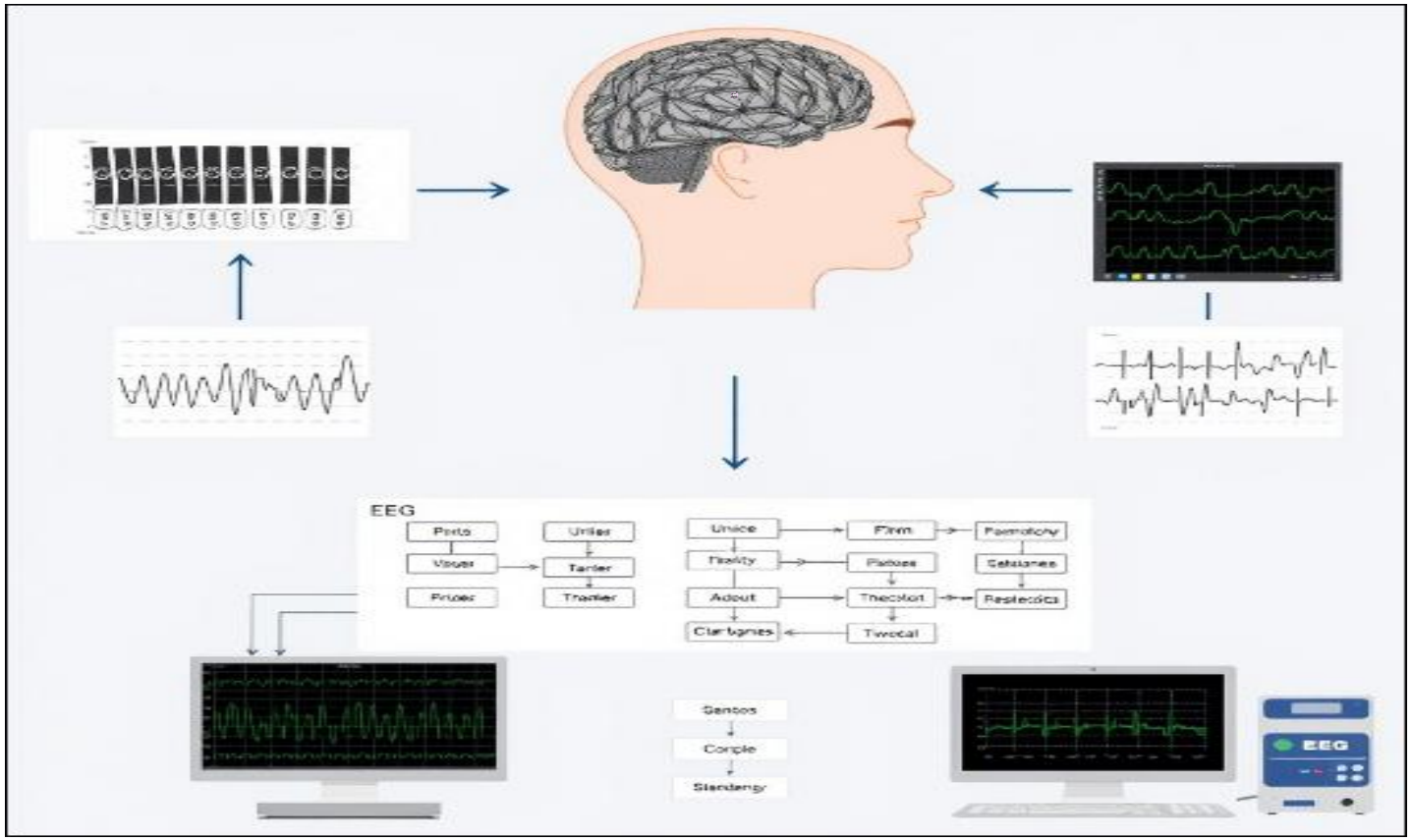
Cour :DSP digital signal processing



Introduction

1. Signal

A physical quantity depending on time or another variable that conveys useful information to the observer.

- **Example:**
- A sound beep → sound.
- Light → light wave.
- Music → acoustic → a piece or a song.
- Photography → image → portraits of a person.

Note: The primary role of the signal is communication in disciplines such as automation, computer science, and electrical engineering.

- **2.The theory of signals** is the set of conceptual tools that allow the construction of theoretical models based on physical laws expressed in mathematical language to provide a representation of the signal.

Introduction

3.DSP

It is the discipline or technique that relies on the resources of electronics, computer science, and applied physics, aimed at the development or interpretation of signals that carry information.

4.Signal Processing in Computer Science

Signal processing in CS involves manipulating, analyzing, and transforming digital or analog signals into actionable information using algorithms and software. This field combines mathematical tools, engineering techniques, and computer science concepts to process a variety of signals, such as audio, video, biomedical signals, or data from sensors.

Signal Processing in Computer Science

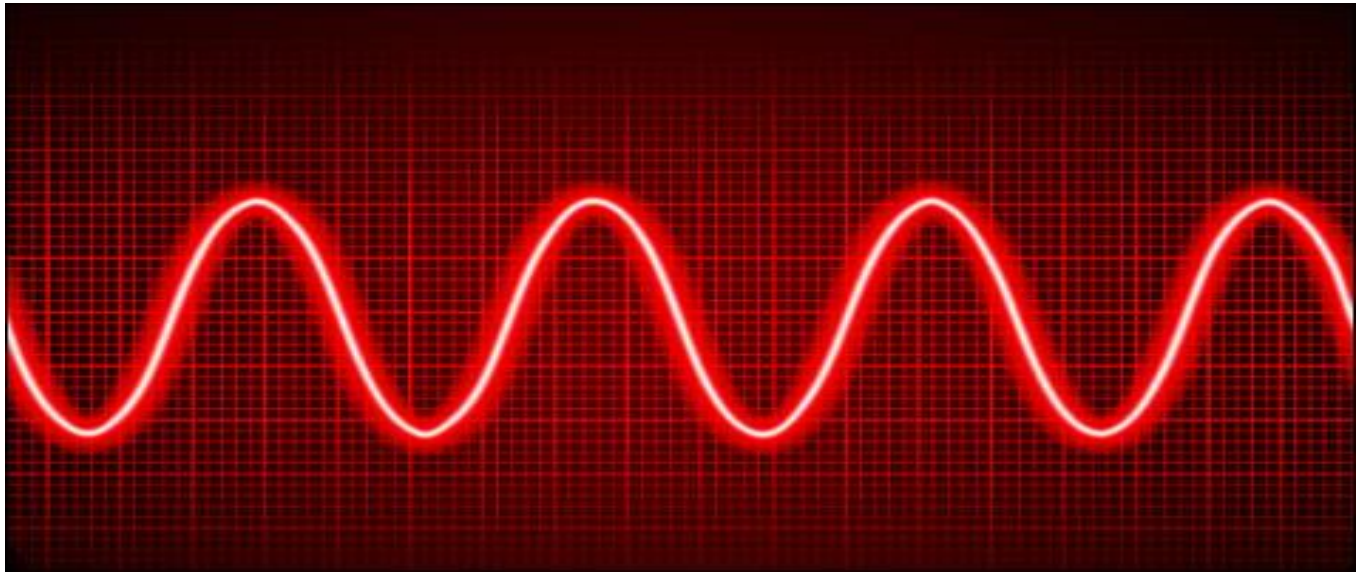
Application Domain	Description
Audio and Speech Processing	Enhancement, speech recognition, speech synthesis, audio compression (MP3, AAC)
Image and Video Processing	Image enhancement, compression (JPEG, MPEG), facial recognition, object tracking
Telecommunications and Networks	Modulation/demodulation, data compression, interference and error reduction
EEG and ECG (Biomedical Signals)	Analysis of biological signals (neurological and cardiac), brain-computer interfaces (BCI)
Control Systems and Robotics	Filtering of sensor signals, control of drones and autonomous vehicles
Radar and Sonar	Detection/localization of objects underwater (sonar) and in the air (radar), radar imaging
Finance and Algorithmic Trading	Analysis of stock market signals, trend detection, fraud prevention
Biomedical Signal Processing	Medical diagnostics, real-time patient monitoring in hospitals

The Time Representation of a Signal

- The signal's medium is most often a voltage but can also be a current, an electromagnetic wave (radio, TV, radar), a light or infrared wave (optical fibers), or a sound wave.
- **The information carried includes:**
  - An audio message (speech, music)
  - A video message (TV image)
  - A binary message (computer-to-printer connection)
  - An analog signal representing the state of a sensor
- A first way to understand a signal is to observe its shape over time as shown by the oscillogram.
- **Signals can thus be classified into four categories:**
  - Periodic signals
  - Non-periodic deterministic signals
  - Pseudo-random signals (audio-video signals)
  - Random signals (electrical noise)

The Time Representation of a Signal

Oscillograms provide information about amplitude, peak value, average value, etc., but not about the frequencies contained in the signal.



Definition 1: The oscillogram of a signal is the representation of the variations in its amplitude as a function of time.

Characteristics of a Sinusoidal Signal

A **sinusoidal signal** is a signal (wave) whose amplitude, observed at a specific location, is a sinusoidal function of time.
The **sine** function is a function that allows calculating the sine of an angle based on the value of that angle.
A sinusoid is the shape that this function takes.

A sinusoidal signal is characterized by its maximum amplitude and its frequency. It can be expressed in the form:
where:

$$s(t) = A\ \sin(\omega t + \varphi)$$

A: Amplitude of the quantity, also called the peak value, in the unit of the measured quantity.
ω: Angular frequency of the quantity in rad·s^{-1}.
ωt + ϕ: Instantaneous phase in radians.
ϕ: Initial phase (or phase at the origin) in radians (often set by the experimenter).

Characteristics of a Sinusoidal Signal

- **Amplitude (A)**:

Represents the maximum value or peak value of the signal.

Measured in the same units as the physical quantity being represented (e.g., volts, meters, etc.).

- **Frequency (f)**:

The number of complete cycles (oscillations) the signal completes per second.

Measured in Hertz (Hz).

- **Angular Frequency (ω)**:

Related to the frequency by the formula:

$\omega = 2\pi f$

Measured in radians per second ($rad \cdot s^{-1}$).

- **Period (T)**:

The time taken to complete one full cycle of the waveform.

Related to the frequency by: $\boxed{T = \frac{1}{f}}$

Measured in seconds (s).